

Entropic Transport Couplings for Gaussian Smoothing and Variance-Preserving Diffusions

Abstract

This note gives a self-contained derivation of an exact entropic optimal transport coupling associated with Gaussian smoothing and with the variance-preserving Ornstein–Uhlenbeck diffusion used in diffusion models. For the quadratic cost $c(x, y) = \frac{1}{2}\|x - y\|^2$, an affine Gaussian channel

$$Y = aX + \sqrt{q} Z, \quad Z \sim \mathcal{N}(0, I_d),$$

is the entropic optimizer between its input and output laws when the entropy parameter is $\varepsilon = q/a$. Gaussian smoothing corresponds to $a = 1$, while a variance-preserving diffusion has $q = 1 - a^2$. We record the Sinkhorn potentials, forward and reverse conditional laws, the score/barycentric formula, the formulation with an OU-adapted cost, arbitrary two-time couplings, and the infinite-time limit.

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1 Conventions and problem formulation

Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ be probability measures with finite second moments. Their set of couplings is denoted by

$$\Pi(\mu, \nu) := \{\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) : (P_1)_\# \pi = \mu, (P_2)_\# \pi = \nu\}.$$

For $\varepsilon > 0$, the entropic optimal transport (EOT) problem considered throughout the note is

$$\inf_{\pi \in \Pi(\mu, \nu)} \left\{ \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2} \|x - y\|^2 d\pi(x, y) + \varepsilon \text{KL}(\pi \mid \mu \otimes \nu) \right\}. \quad (1)$$

Here

$$\text{KL}(\pi \mid r) = \begin{cases} \int \log\left(\frac{d\pi}{dr}\right) d\pi, & \pi \ll r, \\ +\infty, & \text{otherwise.} \end{cases}$$

The normalization of the cost matters. With the alternative cost $\|x - y\|^2$, every matching value of ε below must be multiplied by 2.

Under standard integrability assumptions, the minimizer of (1) is unique and has the Sinkhorn form

$$\frac{d\pi^*}{d(\mu \otimes \nu)}(x, y) = \exp\left(\frac{f(x) + g(y) - \frac{1}{2}\|x - y\|^2}{\varepsilon}\right), \quad (2)$$

where f and g are defined up to the gauge transformation $(f, g) \mapsto (f + C, g - C)$.

Remark 1.1 (Assumptions). The formulas below are identities of measures and remain meaningful for a singular input law μ (for instance, an empirical distribution). To avoid inessential qualifications when manipulating objective values, it is enough to assume that the input has a finite second moment. Convolution with a nondegenerate Gaussian then gives a positive smooth output density and the Gaussian channel has finite mutual information. The endpoint $q = 0$ is understood as a weak limit.

2 The affine Gaussian channel theorem

The following result contains both Gaussian smoothing and the VP diffusion as special cases.

Theorem 2.1 (Affine Gaussian channel as an EOT optimizer). *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, let $a > 0$ and $q > 0$, and define*

$$X \sim \mu, \quad Z \sim \mathcal{N}(0, I_d) \text{ independent}, \quad Y = aX + \sqrt{q}Z. \quad (3)$$

Let $\nu = \text{Law}(Y)$ and let ρ be its Lebesgue density. Then the joint law

$$\pi_{a,q}(dx, dy) = \mu(dx) \mathcal{N}(ax, qI_d)(dy) \quad (4)$$

is the unique minimizer of (1) between μ and ν for

$$\boxed{\varepsilon = \frac{q}{a}}. \quad (5)$$

One choice of Sinkhorn potentials is

$$f(x) = \frac{1-a}{2} \|x\|^2, \quad (6)$$

$$g(y) = -\frac{1-a}{2a} \|y\|^2 - \varepsilon \log \rho(y) - \frac{d\varepsilon}{2} \log(2\pi q). \quad (7)$$

Proof. Because the second marginal of (4) has density ρ ,

$$\frac{d\pi_{a,q}}{d(\mu \otimes \nu)}(x, y) = \frac{(2\pi q)^{-d/2}}{\rho(y)} \exp\left(-\frac{\|y - ax\|^2}{2q}\right). \quad (8)$$

The exponent is separated using

$$-\frac{\|y - ax\|^2}{2q} = -\frac{\|x - y\|^2}{2\varepsilon} + \frac{a(1-a)}{2q}\|x\|^2 - \frac{1-a}{2q}\|y\|^2, \quad (9)$$

where $\varepsilon = q/a$. Substitution into (8) gives the Sinkhorn form (2) with (6)–(7).

For completeness, the optimality can also be seen directly. If $\pi \in \Pi(\mu, \nu)$, then, whenever the quantities are finite,

$$\begin{aligned} \frac{1}{2} \int \|x - y\|^2 d\pi + \varepsilon \text{KL}(\pi \mid \mu \otimes \nu) \\ = \varepsilon \text{KL}(\pi \mid \pi_{a,q}) + C_{\mu,\nu,a,q}, \end{aligned} \quad (10)$$

where the constant is independent of π . Indeed, insert (8); all remaining terms depend only on the fixed first or second marginal. Relative entropy is nonnegative and vanishes only when $\pi = \pi_{a,q}$. A standard truncation argument extends the conclusion when some intermediate terms are infinite. $\square$

Remark 2.2 (Why the matching is forced). The cross term in the log-density of the channel is $a\langle x, y \rangle/q$. The cross term in $-\|x - y\|^2/(2\varepsilon)$ is $\langle x, y \rangle/\varepsilon$. Hence the only possible matching for the standard quadratic cost is $\varepsilon = q/a$.

2.1 An adapted-cost formulation

Define

$$c_a(x, y) := \frac{1}{2}\|y - ax\|^2. \quad (11)$$

The same coupling is equivalently characterized by

$$\pi_{a,q} = \operatorname{argmin}_{\pi \in \Pi(\mu, \nu)} \left\{ \int c_a d\pi q \text{KL}(\pi \mid \mu \otimes \nu) \right\}. \quad (12)$$

Indeed, its potentials for this problem can be chosen as

$$\tilde{f}(x) = 0, \quad \tilde{g}(y) = -q \log \rho(y) - \frac{dq}{2} \log(2\pi q). \quad (13)$$

This formulation makes the reference Gaussian channel completely explicit.

3 Gaussian smoothing

Let

$$\gamma_{\sigma^2} := \mathcal{N}(0, \sigma^2 I_d), \quad \nu = \mu * \gamma_{\sigma^2}.$$

Taking $a = 1$ and $q = \sigma^2$ in Theorem 2.1 gives the following.

Corollary 3.1 (Matched Gaussian smoothing). *For the cost $\frac{1}{2}\|x - y\|^2$, the EOT parameter matched to a Gaussian smoothing of variance σ^2 is*

$$\boxed{\varepsilon = \sigma^2}.$$

The optimal coupling is the additive Gaussian coupling

$$\boxed{\pi_\varepsilon(dx, dy) = \mu(dx) \gamma_\varepsilon(dy - x),} \quad (14)$$

or, probabilistically, $Y = X + \sqrt{\varepsilon}Z$.

If ρ_ε denotes the density of $\mu * \gamma_\varepsilon$, then

$$\frac{d\pi_\varepsilon}{d(\mu \otimes (\mu * \gamma_\varepsilon))}(x, y) = \frac{(2\pi\varepsilon)^{-d/2}}{\rho_\varepsilon(y)} \exp\left(-\frac{\|x - y\|^2}{2\varepsilon}\right). \quad (15)$$

Thus one may take

$$f(x) = 0, \quad g(y) = -\varepsilon \log \rho_\varepsilon(y) - \frac{d\varepsilon}{2} \log(2\pi\varepsilon). \quad (16)$$

The forward and reverse kernels are

$$Y \mid X = x \sim \mathcal{N}(x, \varepsilon I_d), \quad (17)$$

$$\pi_\varepsilon(dx \mid Y = y) = \frac{\gamma_\varepsilon(y - x)\mu(dx)}{(\mu * \gamma_\varepsilon)(y)}. \quad (18)$$

Differentiating the Gaussian convolution yields Tweedie's identity

$$\boxed{\mathbb{E}[X \mid Y = y] = y + \varepsilon \nabla \log \rho_\varepsilon(y).} \quad (19)$$

If the heat semigroup is normalized as $P_t\mu = \mu * \mathcal{N}(0, 2tI_d)$, the matching is $\varepsilon = 2t$. If the cost is $\|x - y\|^2$ instead of half that quantity, the matching becomes $\varepsilon = 2\sigma^2$.

Remark 3.2 (Unmatched parameters). For a general non-Gaussian μ , the additive Gaussian coupling is not the EOT optimizer when $\varepsilon \neq \sigma^2$ under our cost convention. The cross-term test in the preceding remark already shows the obstruction. The unmatched problem generally requires two nontrivial Sinkhorn potentials. When both marginals are Gaussian, the optimizer is itself Gaussian and a separate closed-form covariance calculation is possible.

4 Variance-preserving diffusion

4.1 The forward SDE and its marginals

Let $\beta : [0, \infty) \rightarrow [0, \infty)$ be locally integrable and consider the variance-preserving (VP) SDE

$$dX_t = -\frac{\beta(t)}{2}X_t dt + \sqrt{\beta(t)} dW_t, \quad X_0 \sim \mu_0. \quad (20)$$

Its Fokker–Planck equation is

$$\partial_t \rho_t = \frac{\beta(t)}{2} \nabla \cdot (x \rho_t) + \frac{\beta(t)}{2} \Delta \rho_t. \quad (21)$$

After the time change $\tau(t) = \frac{1}{2} \int_0^t \beta(r) dr$, this is the Ornstein–Uhlenbeck equation associated with the quadratic potential $V(x) = \frac{1}{2} \|x\|^2$.

Define

$$B_t := \int_0^t \beta(r) dr, \quad a_t := e^{-B_t/2}, \quad q_t := 1 - a_t^2. \quad (22)$$

Solving the linear SDE gives

$$X_t = a_t X_0 + \sqrt{q_t} Z, \quad Z \sim \mathcal{N}(0, I_d) \text{ independent of } X_0, \quad (23)$$

and hence

$$\mu_t = (a_t \text{Id})_\# \mu_0 * \mathcal{N}(0, q_t I_d). \quad (24)$$

The identity $q_t = 1 - a_t^2$ is the variance-preserving property: if $X_0 \sim \mathcal{N}(0, I_d)$, then $X_t \sim \mathcal{N}(0, I_d)$ at every time.

4.2 The exact finite-time Sinkhorn coupling

Suppose $t > 0$ and $B_t > 0$, so $0 < a_t < 1$ and $q_t > 0$. The pathwise forward coupling between μ_0 and μ_t is

$$\pi_t(\mathrm{d}x, \mathrm{d}y) = \mu_0(\mathrm{d}x) \mathcal{N}(a_t x, q_t I_d)(\mathrm{d}y). \quad (25)$$

Theorem 2.1, with $a = a_t$ and $q = q_t$, shows that this is exactly the standard quadratic EOT coupling for

$$\varepsilon_t = \frac{q_t}{a_t} = \frac{1 - a_t^2}{a_t} = a_t^{-1} - a_t = 2 \sinh\left(\frac{B_t}{2}\right). \quad (26)$$

Writing ρ_t for the density of μ_t , its density ratio is

$$\frac{\mathrm{d}\pi_t}{\mathrm{d}(\mu_0 \otimes \mu_t)}(x, y) = \frac{(2\pi q_t)^{-d/2}}{\rho_t(y)} \exp\left(-\frac{\|y - a_t x\|^2}{2q_t}\right), \quad (27)$$

and one may choose the potentials

$$f_t(x) = \frac{1 - a_t}{2} \|x\|^2, \quad (28)$$

$$g_t(y) = -\frac{1 - a_t}{2a_t} \|y\|^2 - \varepsilon_t \log \rho_t(y) - \frac{d\varepsilon_t}{2} \log(2\pi q_t). \quad (29)$$

At $t = 0$, $a_0 = 1$, $q_0 = 0$, and $\varepsilon_0 = 0$. The coupling is the diagonal law $(\mathrm{Id}, \mathrm{Id})_{\#} \mu_0$. This endpoint is a zero-noise limit, not an entropy-regularized problem with positive ε .

4.3 OU-adapted cost

For the time-dependent cost

$$c_t^{\mathrm{OU}}(x, y) := \frac{1}{2} \|y - a_t x\|^2, \quad (30)$$

the same coupling satisfies

$$\pi_t = \operatorname{argmin}_{\pi \in \Pi(\mu_0, \mu_t)} \left\{ \int c_t^{\mathrm{OU}} \mathrm{d}\pi \, \mathrm{KL}(\pi \mid \mu_0 \otimes \mu_t) \right\}. \quad (31)$$

The potentials simplify to

$$\tilde{f}_t(x) = 0, \quad \tilde{g}_t(y) = -q_t \log \rho_t(y) - \frac{dq_t}{2} \log(2\pi q_t). \quad (32)$$

This is the most direct one-step Schrödinger formulation: the Gibbs kernel is precisely the OU transition kernel. Since that reference process is initialized at μ_0 , its endpoint law is already μ_t , and its joint endpoint law is therefore feasible and minimizes the relative entropy.

4.4 Reverse kernel and denoising formula

The forward conditional law is

$$X_t \mid X_0 = x \sim \mathcal{N}(a_t x, q_t I_d).$$

Bayes' formula gives the reverse conditional law

$$\pi_t(\mathrm{d}x \mid X_t = y) = \frac{\exp\left(-\frac{\|y - a_t x\|^2}{2q_t}\right) \mu_0(\mathrm{d}x)}{\int \exp\left(-\frac{\|y - a_t z\|^2}{2q_t}\right) \mu_0(\mathrm{d}z)}. \quad (33)$$

Differentiation under the integral in (24) yields

$$\nabla \log \rho_t(y) = -\frac{1}{q_t} (y - a_t \mathbb{E}[X_0 \mid X_t = y]). \quad (34)$$

Consequently, the reverse barycentric projection is

$$\boxed{\mathbb{E}[X_0 \mid X_t = y] = \frac{1}{a_t} (y + q_t \nabla \log \rho_t(y))}. \quad (35)$$

This is the VP version of Tweedie's formula and is the precise relation between the entropic coupling, denoising, and the score of the noised data distribution.

4.5 Any two diffusion times

The same result holds between arbitrary times $0 \leq s < t$. Set

$$a_{s,t} := \exp\left(-\frac{1}{2} \int_s^t \beta(r) \, dr\right), \quad q_{s,t} := 1 - a_{s,t}^2. \quad (36)$$

The Markov transition is

$$X_t = a_{s,t} X_s + \sqrt{q_{s,t}} Z_{s,t}, \quad Z_{s,t} \sim \mathcal{N}(0, I_d), \quad Z_{s,t} \perp X_s.$$

Therefore

$$\pi_{s,t}(\mathrm{d}x, \mathrm{d}y) = \mu_s(\mathrm{d}x) \mathcal{N}(a_{s,t}x, q_{s,t}I_d)(\mathrm{d}y) \quad (37)$$

is the exact standard quadratic Sinkhorn coupling between μ_s and μ_t with

$$\boxed{\varepsilon_{s,t} = \frac{q_{s,t}}{a_{s,t}} = 2 \sinh\left(\frac{1}{2} \int_s^t \beta(r) \, dr\right)}. \quad (38)$$

All preceding formulas apply after replacing (μ_0, μ_t, a_t, q_t) with $(\mu_s, \mu_t, a_{s,t}, q_{s,t})$.

4.6 Common time normalizations

If $\beta(t) \equiv \beta$ is constant, then

$$a_t = e^{-\beta t/2}, \quad q_t = 1 - e^{-\beta t}, \quad \varepsilon_t = 2 \sinh(\beta t/2).$$

For the common OU normalization

$$\mathrm{d}X_t = -X_t \mathrm{d}t + \sqrt{2} \mathrm{d}W_t, \quad (39)$$

one has $\beta = 2$ and hence

$$a_t = e^{-t}, \quad q_t = 1 - e^{-2t}, \quad \boxed{\varepsilon_t = 2 \sinh t}. \quad (40)$$

5 The infinite-time limit

Assume

$$\int_0^\infty \beta(r) \, dr = +\infty. \quad (41)$$

Then $a_t \rightarrow 0$, $q_t \rightarrow 1$, and

$$\mu_t \implies \gamma := \mathcal{N}(0, I_d). \quad (42)$$

The forward endpoint coupling converges weakly to

$$\boxed{\pi_\infty(\mathrm{d}x, \mathrm{d}y) = \mu_0(\mathrm{d}x)\gamma(\mathrm{d}y) = \mu_0 \otimes \gamma.} \quad (43)$$

Thus the initial state and the stationary endpoint are independent.

For the standard quadratic cost,

$$\varepsilon_t = \frac{1 - a_t^2}{a_t} \longrightarrow +\infty. \quad (44)$$

Accordingly, (43) is the infinite-regularization limit of the standard quadratic Sinkhorn coupling. Except in degenerate cases (for example, a deterministic first marginal), the product coupling is not the optimizer for any finite ε : at finite regularization the quadratic Gibbs kernel retains a nonzero x - y interaction.

For the OU-adapted formulation,

$$c_\infty^{\text{OU}}(x, y) = \frac{1}{2}\|y\|^2, \quad q_\infty = 1.$$

The cost depends only on the fixed second marginal and is therefore constant over $\Pi(\mu_0, \gamma)$. It follows directly that

$$\pi_\infty = \underset{\pi \in \Pi(\mu_0, \gamma)}{\operatorname{argmin}} \operatorname{KL}(\pi \mid \mu_0 \otimes \gamma) = \mu_0 \otimes \gamma. \quad (45)$$

Unlike the standard-cost parameter, the adapted entropy parameter has the finite limit $q_\infty = 1$; the interaction instead disappears from the cost because $a_t \rightarrow 0$.

6 Objective value and mutual information

The KL term of the forward coupling has a direct information-theoretic meaning:

$$\operatorname{KL}(\pi_{a,q} \mid \mu \otimes \nu) = I(X; Y). \quad (46)$$

Moreover,

$$\mathbb{E}\|X - Y\|^2 = (1 - a)^2 \mathbb{E}\|X\|^2 + dq. \quad (47)$$

Hence the minimum value of the standard-cost problem in Theorem 2.1 is

$$\frac{1}{2} \left((1 - a)^2 \mathbb{E}\|X\|^2 + dq \right) + \frac{q}{a} I(X; Y). \quad (48)$$

When differential entropies are finite,

$$I(X; Y) = h(Y) - h(Y \mid X) = h(\nu) - \frac{d}{2} \log(2\pi e q). \quad (49)$$

For the VP diffusion, substitute $(a, q) = (a_t, q_t)$ in (48). As $t \rightarrow \infty$, the mutual information tends to zero under standard moment assumptions, consistently with the limiting product coupling.

Setting	Cost and entropy term	Matching parameter
Gaussian smoothing	$\frac{1}{2}\ x - y\ ^2 + \varepsilon \text{KL}$	$\varepsilon = \sigma^2$
Gaussian smoothing	$\ x - y\ ^2 + \varepsilon \text{KL}$	$\varepsilon = 2\sigma^2$
VP, finite t	$\frac{1}{2}\ x - y\ ^2 + \varepsilon_t \text{KL}$	$\varepsilon_t = (1 - a_t^2)/a_t$
VP, finite t	$\frac{1}{2}\ y - a_t x\ ^2 + q_t \text{KL}$	$q_t = 1 - a_t^2$
VP, $t = \infty$	standard quadratic cost	$\varepsilon_t \rightarrow +\infty$
VP, $t = \infty$	$\frac{1}{2}\ y\ ^2 + \text{KL}$	product coupling

Table 1: The entropy coefficient under the cost convention used in this note. Here KL abbreviates $\text{KL}(\pi \mid \text{product of marginals})$.

7 Summary of parameter conventions

The central identities are therefore

$$\begin{aligned}
 X_t &= a_t X_0 + \sqrt{1 - a_t^2} Z, \\
 \pi_t &= \text{Law}(X_0, X_t), \\
 \varepsilon_t &= \frac{1 - a_t^2}{a_t}, \\
 \mathbb{E}[X_0 \mid X_t = y] &= a_t^{-1}(y + (1 - a_t^2)\nabla \log \rho_t(y)), \\
 \pi_t &\implies \mu_0 \otimes \mathcal{N}(0, I_d) \quad \text{as } t \rightarrow \infty.
 \end{aligned} \tag{50}$$

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